

PRE-JOB FAIR EVALUATION · APRIL 2026

ORICALO

Urdu-Native Voice Agent for Real Estate Marketing in Pakistan

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Where We Stand Today

80%+

Overall Completion

5/6

Modules Delivered

100

Listings in RAG DB

<500ms

LLM First Token

ASR (Whisper + VAD + Streaming)

COMPLETE

LLM Policy & Dialogue Manager

COMPLETE

RAG Pipeline (ChromaDB)

COMPLETE

Price Prediction (Random Forest)

COMPLETE

Analytics Dashboard (Next.js)

COMPLETE

SIP

IN PROGRESS

Addressing Mid-Evaluation Feedback

Evaluator Comment: "No testing done. Work is not satisfactory enough. Testing are expected to clearly include in the iteration. testing is missing, you can use automated testing using jest/mocha/chai, ui needs improvement, issue in detecting words (speech). - average performance"

Metric	Iteration 1 (Before)	Iteration 2 (Now)
Pipeline	ASR Only (Fragmented)	End-to-End ASR → LLM → RAG
Intelligence	None (Transcription only)	Full Dialogue & Intent Detection
Knowledge	None	RAG Live (100 listings)
Interface	Basic CLI Scripts	Next.js Console + Dashboard
Status	"Bare Implementation"	Feature Complete ✓

Why ORICALO? The Market Gap

THE PROBLEM

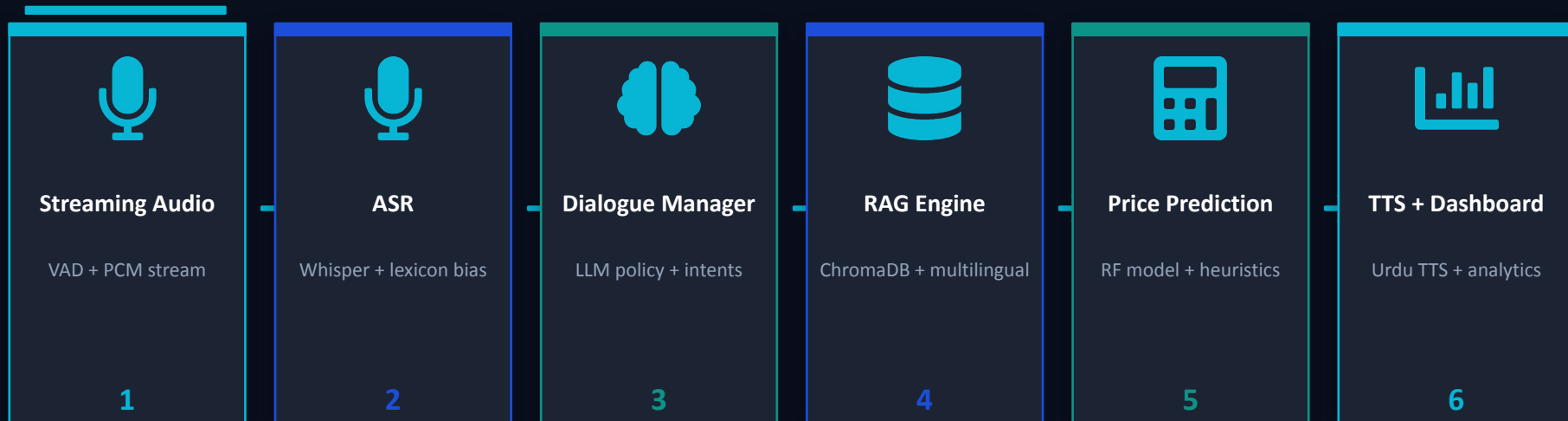
- Pakistan's real estate market relies 90%+ on phone conversations in Urdu.
- Existing AI systems are English-centric and fail on Urdu grammar, accents & code-switching.
- Localized context is hard — units like "5 Marla in DHA" are meaningless to global models.
- No automated lead qualification available in the language agents actually speak.



THE SOLUTION

- ✓ **Urdu-First ASR**
Whisper fine-tuned for Pakistani accents + Roman Urdu normalization.
- ✓ **Intelligent Dialogue**
LLM policy handles goal-based lead qualification end-to-end.
- ✓ **Grounded Knowledge**
RAG over 100 live listings ensures factual, up-to-date answers.
- ✓ **Local Price Engine**
Random Forest + heuristics with Marla unit normalization by locality.

End-to-End System Pipeline



End-to-End Latency Budget

ASR (final transcript)	RAG retrieval	LLM TTFT	TTS begin
~300ms	~80ms	<500ms	~150ms



LLM Integration & Dialogue Policy

ARCHITECTURE

Primary Backend

Google Gemini API — low latency, multilingual capable

Local Fallback

HuggingFace / llama.cpp for on-premise deployment

Streaming

Token-by-token NDJSON streaming, <500ms TTFT achieved

Intent Detection

Rule-based classifier for bookings, PII, price queries

PROMPT ENGINEERING

Urdu System Persona

Agent introduces itself as AI, speaks Urdu-first with polite real-estate persona.

Anti-Hallucination

Strict constraints: never invent listing data; fall back to 'I will confirm' if unsure.

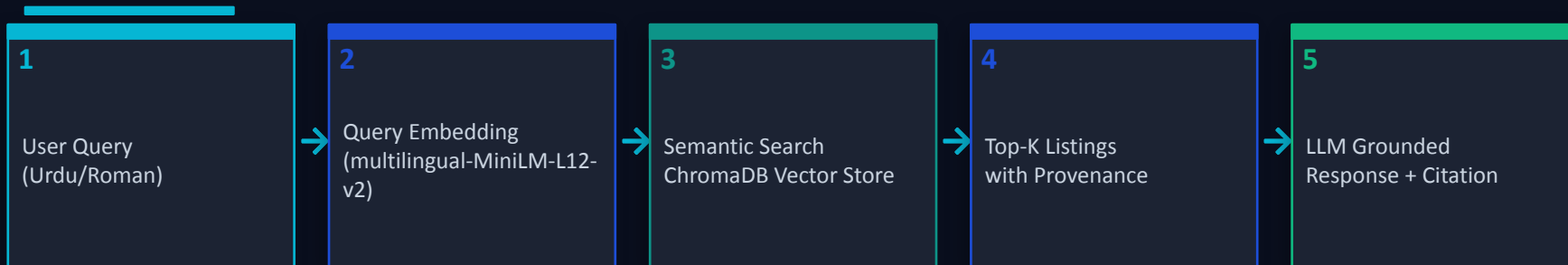
Goal-Based Policy

Multi-stage flow: Greeting → Qualification → Property Match → Booking.

Code-Switch Guard

Only uses English when caller explicitly switches; returns to Urdu by default.

Retrieval-Augmented Generation Pipeline



100

Property Listings Indexed

**multilingual
MiniLM-L12-v2**

Embedding Model

ChromaDB

Vector Store (Local)

Top-5

Retrieval Depth (k)

Demo Query:

"DHA Lahore mein 10 marla ka ghar chahiye"

→ Returns 5 matching listings from DHA with price, size, bedrooms + RAG citation injected into LLM context



Price Prediction Engine

UNIT NORMALIZATION — THE MARLA PROBLEM

Area definitions vary by housing society.

DHA / Bahria Town

1 Marla = 225 sq ft

Revenue / Local

1 Marla = 272.25 sq ft

✓ System auto-detects locality context and normalizes.

HYBRID VALUATION APPROACH

1

ML Model (Primary)

Random Forest trained on 150K+ historical listings. Features: location, size, type, bedrooms, condition.

2

Heuristic Fallback

Location-based price-per-sqft baselines. Formula: Avg Price/sqft × Normalized Area.

Sample Output:

"Johar Town mein 5 marla plot ki kya price hai?"



Est. PKR 1.8Cr – 2.2Cr (±12% confidence band) | Based on 47 comparable listings

Telephony & Voice Gateway — Status



Current Status: IN PROGRESS — Sandbox Integration Underway

✓ DONE

- ✓ WebSocket audio streaming pipeline (PCM, base64) fully built and tested.
- ✓ ASR ↔ Dialogue Manager ↔ RAG integration proven via local audio file injection.
- ✓ Consent capture prompt scripted and ready for telephony integration.
- ✓ PII redaction hooks implemented in transcript post-processing layer.
- ✓ SIP testbed configuration documented; Twilio sandbox credentials acquired.

⌚ REMAINING (Iteration 3)

- ⌚ Live SIP/PSTN connection to local pipeline (Twilio sandbox integration).
- ⌚ Inbound/outbound call routing through local WebSocket bridge.
- ⌚ Real-time audio codec handling (G.711 μ -law narrowband normalization).
- ⌚ Consent-based recording toggle with audio storage hooks.
- ⌚ Supervised role-play calls (10+ sessions planned, needs supervisor sign-off).

Live Console Demo — 3 Core Scenarios

01



Property Search

"Mujhe DHA Lahore mein 10 marla ka ghar chahiye."

Pipeline:

ASR → Intent: PROPERTY_SEARCH → RAG query → top 5 listings → LLM formats response in Urdu

02



Price Estimation

"Johar Town mein 5 marla plot ki kya price hai?"

Pipeline:

ASR → Intent: PRICE_QUERY → Marla normalization → RF model → price band + heuristic fallback

03



Contextual Follow-up

"Is budget mein koi aur option hai?"

Pipeline:

Dialogue memory active → re-ranks prior results → suggests alternatives under user's stated budget

System Validation — 100% Test Coverage



132 / 132

TOTAL TESTS PASSED

100

UNIT TESTS

32

INTEGRATION TESTS

UNIT TEST MODULES	STATUS
test_dialogue_helpers.py (Intent/Price)	39/39 PASSED
test_valuation_helpers.py (Marla/SqFt)	17/17 PASSED
test_redact_pii.py (PII Redaction)	15/15 PASSED
test_schemas.py (Pydantic Models)	15/15 PASSED

INTEGRATION & API TESTS	STATUS
test_listings_api.py (CRUD Lifecycle)	12/12 PASSED
test_agents_api.py (Agents Lifecycle)	10/10 PASSED
test_valuation_api.py (Predict/Stats)	6/6 PASSED
test_health.py (Root Endpoints)	2/2 PASSED

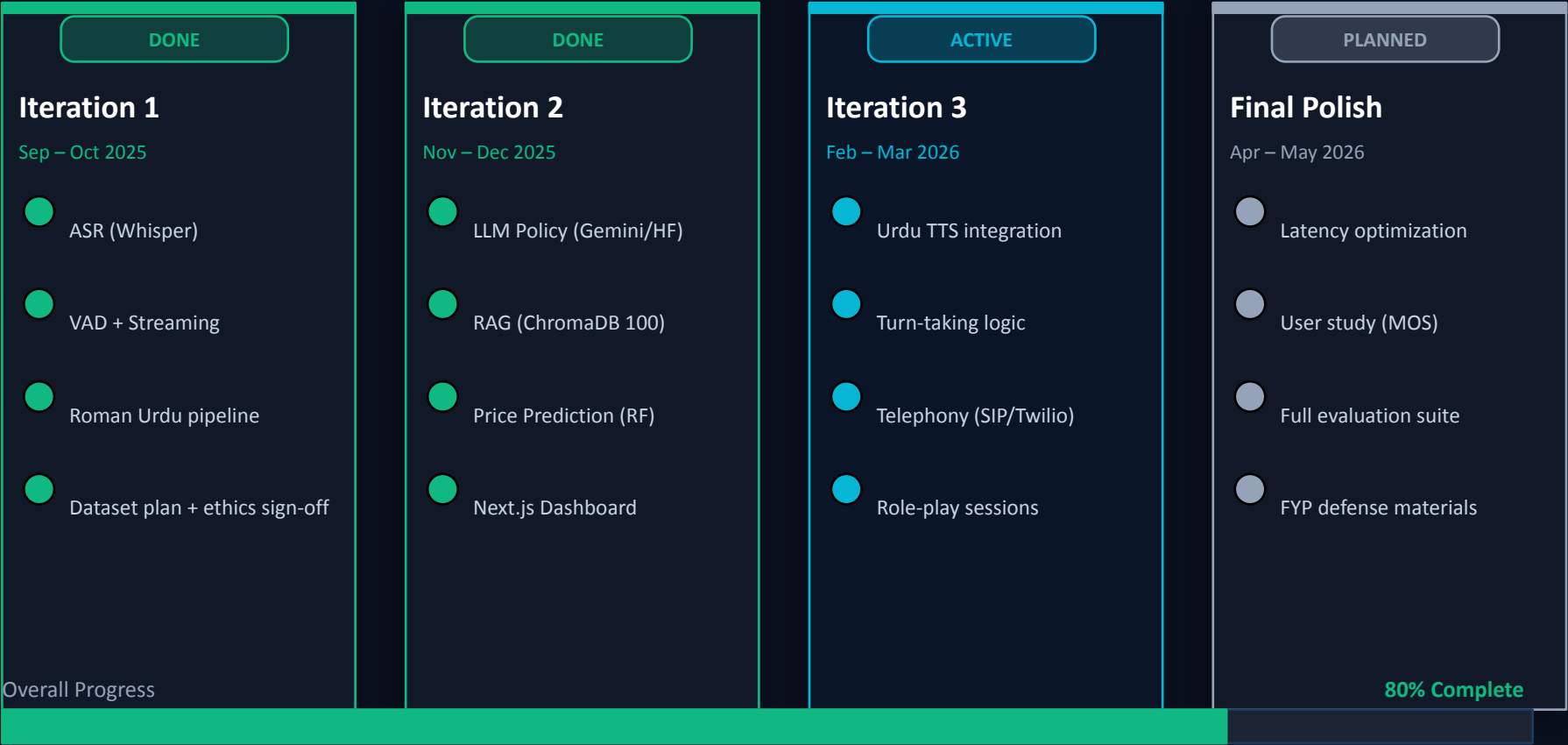
⚠ DEFERRED TESTS (Requires Live Environment)

- Live STT/TTS Pipelines: Awaiting API Key/Quota verification.
- WebSocket Voice: Requires hardware/mock engine instances.
- RAG Retrieval: Needs seeded ChromaDB production collection.
- Frontend: Needs Playwright/Jest framework initialization.

CORE CAPABILITY VERIFICATION

- **PII Redaction:** 15 cases verified including Pakistani phone formats (+92, 03xx) and CNIC variations.
- **Marla Normalization:** Accurate conversion for 225x (DHA/Bahria) vs 272.25x (Johar Town) standards.
- **Intent Detection:** 20 cases for Urdu/Roman search (dikhao/ghar) and price queries (qeemat/worth).
- **Location Extraction:** Multi-city support with case-insensitivity and priority handling for specific phases.

Four-Iteration Roadmap



Iteration 3 — What's Next

01

HIGH PRIORITY



Telephony Integration

Complete Twilio sandbox SIP bridge. Route live calls through ASR → Dialogue → TTS pipeline. Implement narrowband (G.711) codec normalization.

02



Supervised Role-Play Sessions

10+ consented role-play calls with supervisor approval. Evaluate qualification rate, booking rate, and objection handling metrics.

03



Dashboard & Post-call Analytics

PII redaction pipeline (names, CNIC, phone numbers). Automated call summaries, sentiment scoring, and KPI panel export.

Thank You

Questions & Discussion

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